

The size of linearly independent sets

Theorem

If $v_1, \dots, v_m$ are in R^n and $m > n$ then $v_1, \dots, v_m$ are linearly dependent.

If $v_1, \dots, v_m$ are linearly dependent then at least one of these vectors can be written as a linear combination of the others.

$$n \updownarrow \left(\begin{array}{c|c|c|c|c} | & | & | & \dots & | \\ v_1 & v_2 & v_3 & \dots & v_m \\ | & | & | & \dots & | \end{array} \right) \leftarrow n \times m, \text{ call it } A.$$

$$\begin{array}{c} \text{rref} \left\{ \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right. \end{array} \quad \begin{array}{c} \xleftarrow{m} \\ m \end{array} \quad \left(\begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right)$$

leading ones in
the rref but
there must be
some column
without a leading
1 if $m > n$

Some variable in $Ax = 0$ will be free.

So if $v_1, \dots, v_m$ are lin indep in $\mathbb{R}^n$
then $m \leq n$.

Ex: $\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \dots \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$
 $\underline{e_1} \quad \underline{e_2} \quad \underline{e_3} \quad \dots \quad \underline{e_n}$

$$\begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{so } e_1 \dots e_n \text{ are}$$

lin indep and
they span $\mathbb{R}^n$.

Ex

$$\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$v_1 \quad v_2 \quad v_n$

$$\begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix} x = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

The only solution
is $x = 0$.

So $v_1, \dots, v_n$

are lin. indep.

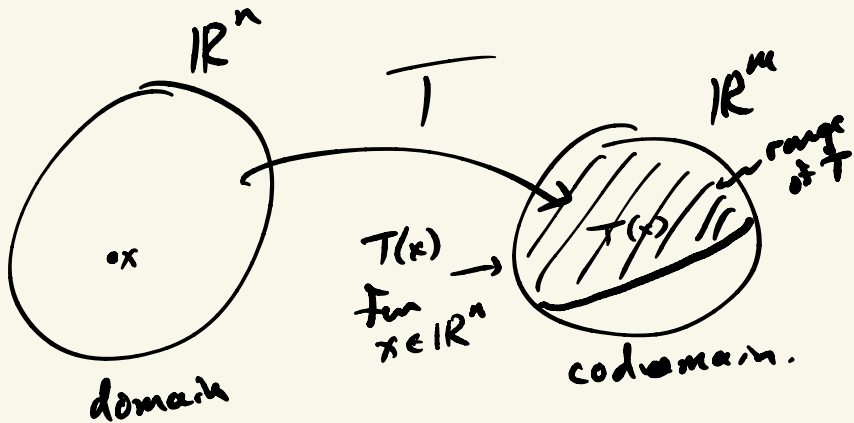
Ex: Show $v_1, \dots, v_n$ spans
 $\mathbb{R}^n$.

Transformations

Definition

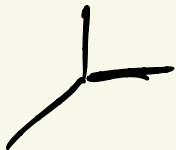
- A **transformation** from R^n to R^m is a function which assigns to every vector in R^n some vector in R^m .
- If this function is T then we write $T : \underline{R^n} \rightarrow \underline{R^m}$.
- R^n is called the domain of T and R^m is the codomain.
- The range of T is the set of all things of the form $T(x)$ for $x \in R^n$.

The most important transformations for us are matrix transformations. Given an $m \times n$ matrix A then we can define a transformation T_A from R^n to R^m by $x \mapsto Ax$, multiplication of a vector in R^n on the left by A .



Ex: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$



$\mathbb{R}^3$
domain.



$\mathbb{R}^3$ is the codomain.
by the xy -plane is
the range

Ex: $T: \mathbb{R}^3 \rightarrow \underline{\mathbb{R}}$.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \sin(x+y+z).$$

domain is $\mathbb{R}^3$

codomain is $\mathbb{R}$

range is $[-1, 1]$.

If A is an $n \times n$ matrix then we define $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ by

$$T_A(x) = Ax \text{ for } x \in \mathbb{R}^n.$$

Ex: $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ $T_A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

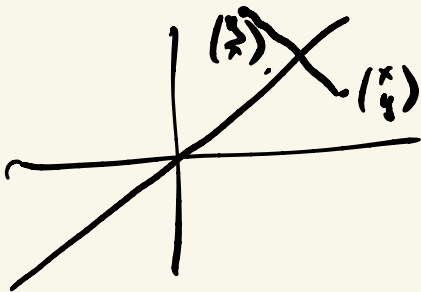
$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

In general T_A is called a matrix transformation.

Ex:

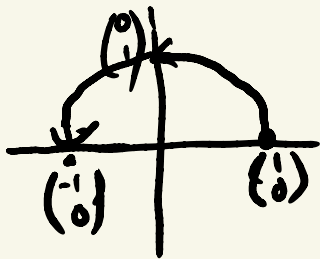
$$\underset{\substack{\uparrow \\ 2 \times 2}}{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T_A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}.$$



Ex: $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$



Rotation counter
clockwise by 90° .

Ex: $A = \begin{pmatrix} 4 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix}$ $T_A: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

$\uparrow$
 3×2

$\mathbb{R}^2$ is the domain, $\mathbb{R}^3$ is the codomain.

The range is the span of the columns of A , in this case $\begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$.

Is $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ in the range of T_A ?

i.e. does $\begin{pmatrix} 4 & 0 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ have a

solution?

$$\begin{pmatrix} 4 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 2 & 1 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

No solution so $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is not in the range of T_A .

Ex:

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 3 & -1 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

4×3

Is $\begin{pmatrix} -1 \\ 6 \\ 1 \\ 2 \end{pmatrix}$ in
the range of T_A ?

$$\begin{pmatrix} 1 & 0 & 2 & -1 \\ 2 & 3 & -1 & 6 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 \end{pmatrix}$$

ref
 $\leadsto$

$$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$x_1 = 1$$

$$x_2 = 1$$

$$x_3 = -1$$

Definition

We say that a transformation $T : R^n \rightarrow R^m$ is a linear transformation if

- for all $u, v \in R^n$, $T(u + v) = T(u) + T(v)$, and
- for all scalars λ and $u \in R^n$, $T(\lambda u) = \lambda T(u)$.

Fact: If T is a linear transformation then $T(0) = 0$ and for all scalars c, d and vectors u, v , $T(cu + dv) = cT(u) + dT(v)$.